

Multi-objective Wedding Planner

- Find all the solutions where no other solution exist with a higher aggregated (summed) shared interest **and** a lower sum of gender balance plus guests who know too few.

Problem 1

We will only show the objective function, since this is the only thing which changes. But we want to maximize the shared interests, and minimize gender plus known guests. So we multiply with -1 on the gender plus know objective.

Maximize. $\sum_{g1,g2,t|g1 < g2} SharedInterests_{g1,g2} \cdot y_{g1,g2,t}$

$$- \sum_t (m_t + f_t) - \sum_g k_g$$

Subject to :

...thesameconstraintsasinthesingle - objectivemodel

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# WeddingPlanner Assignment5 , "Mathematical Programming Modelling" (42112)
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
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# PARAMETERS
include("WeddingData20.jl") # small dataset
println("Runing WeddingPlanner with $(G) guests, $(T) tables with capacity $(TableCap)")
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#####
# Model
mo_wp = Model()

# 1 if guest g is sitting at table T
@variable(mo_wp, x[g=1:G,t=1:T], Bin)

# 1 if guest g1 and guest g2 are both sitting at table T, symmetric

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@variable(mo_wp, 0 <= y[g1=1:G,g2=1:G,t=1:T] <= ( g1 < g2 ? 1 : 0))

# excess males at table t
@variable(mo_wp, m[t=1:T] >= 0)

# excess females at table t
@variable(mo_wp, f[t=1:T] >= 0)

# guest g is missing known people
@variable(mo_wp, k[g=1:G] >= 0)

# Objective 1: Maximize shared interests
@expression(mo_wp, Share_expr, sum( SharedInterests[g1,g2]*y[g1,g2,t]
                                   for t=1:T, g1=1:G, g2=1:G if g1<g2))

# Objective 2: Maximize negative gender and negative friends know
@expression(mo_wp, Gender_Friends_expr,
            - sum( m[t] + f[t] for t=1:T)
            - sum( k[g] for g=1:G))

# Define combined BI-OBJECTIVE
@objective(mo_wp, Max, [Share_expr, Gender_Friends_expr])

# all guests has to sit at a table
@constraint(mo_wp, [g=1:G],
            sum( x[g,t] for t=1:T) == 1)

# dont exceed the number of persons at a table
@constraint(mo_wp, [t=1:T],
            sum( x[g,t] for g=1:G) <= TableCap)

# couples should sit at the same table
@constraint(mo_wp, [g1=1:G,g2=1:G,t=1:T; Couple[g1,g2]==1],
            x[g1,t] == x[g2,t])

# if there are too many males at at table, penalty
@constraint(mo_wp, [t=1:T],
            sum( Male[g]*x[g,t] - Female[g]*x[g,t] for g=1:G) <=
            m[t] + 2)

# if there are too many females at at table, penalty
@constraint(mo_wp, [t=1:T],
            sum( Female[g]*x[g,t] - Male[g]*x[g,t] for g=1:G) <=
            f[t] + 2)

# limit y, can only become 1 if g1 and g2 are at the same table
@constraint(mo_wp, [g1=1:G,g2=1:G,t=1:T; g1<g2],

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        y[g1,g2,t] <= x[g1,t])

# limit y, can only become 1 if g1 and g2 are at the same table
@constraint(mo_wp, [g1=1:G,g2=1:G,t=1:T; g1<g2],
    y[g1,g2,t] <= x[g2,t])

# if a person sits at a table, penalty if not enough known people
@constraint(mo_wp, [g=1:G,t=1:T],
    k[g] - 3*x[g,t] +
    sum( Know[g,g1]*(y[g,g1,t]+y[g1,g,t]) for g1=1:G) >= 0)
#####

#####
# Solve

# Set MIP solver
set_optimizer(mo_wp, () -> MultiObjectiveAlgorithms.Optimizer(HiGHS.Optimizer))
set_silent(mo_wp)

# Set MO solver
# EpsilonConstraint, Works
set_attribute(mo_wp, MultiObjectiveAlgorithms.Algorithm(), MultiObjectiveAlgorithms.Epsilon)

optimize!(mo_wp)
#solution_summary(mo_wp)
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# Solution
if result_count(mo_wp)>=1
    println("No Pareto optimal points: ",result_count(mo_wp))
    for i in 1:result_count(mo_wp)
        obj_vec = objective_value(mo_wp; result = i)
        println("Objective 1: ",round.(Int,obj_vec[1]), "    Objective 2: ",round.(Int,obj_vec[2]))
    end
else
    println("No solutions found")
end
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